

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

MATHEMATICS -- PAPER 1

Duration: 1 Hour | Level: JHS 3 / BECE | Code: MATH 01/02 - P1

INSTRUCTIONS TO CANDIDATES

Answer all 40 questions. Each question is followed by four options lettered A to D. Find the correct option for each question and mark it on your answer sheet. Mathematical tables and graph papers are provided where required.

PART I: OBJECTIVE QUESTIONS

1. If set $P = \{1, 2, 3, 4, 6, 12\}$ and set $Q = \{2, 3, 5, 7, 11\}$, find $P \cap Q$.
 - A. $\{2, 3\}$
 - B. $\{1, 2, 3\}$
 - C. $\{2, 3, 5\}$
 - D. $\{1, 2, 3, 4, 6, 12\}$
2. Simplify: $2^3 \times 3^2 \div 6$.
 - A. 6
 - B. 8
 - C. 12
 - D. 18
3. Express 0.00345 in standard form (scientific notation).
 - A. 3.45×10^{-3}
 - B. 3.45×10^{-2}
 - C. 34.5×10^{-4}
 - D. 3.45×10^3
4. Find the Highest Common Factor (HCF) of 18, 24 and 36.
 - A. 3
 - B. 6
 - C. 12
 - D. 72
5. Evaluate: $\frac{3}{4} - (\frac{1}{2} + \frac{1}{8})$.
 - A. $\frac{1}{8}$
 - B. $\frac{1}{4}$
 - C. $\frac{3}{8}$
 - D. $\frac{1}{2}$
6. Solve for x: $3(x - 2) = 2(x + 4)$.
 - A. 10
 - B. 12
 - C. 14
 - D. 16
7. If 4 men can weed a farm in 6 days, how many days will 3 men take working at the same rate?
 - A. 4.5 days
 - B. 7 days
 - C. 8 days
 - D. 9 days
8. Find the median of the following numbers: 7, 4, 9, 3, 6, 8, 4.
 - A. 4
 - B. 5
 - C. 6
 - D. 7

9. Calculate the simple interest on GHS 12,000 for 3 years at 5% per annum.

- A. GHS 1,200
- B. GHS 1,500
- C. GHS 1,800
- D. GHS 2,400

10. The interior angle of a regular polygon is 120 deg. How many sides does it have?

- A. 5
- B. 6
- C. 7
- D. 8

11. Convert 11011_{two} to a base ten number.

- A. 23
- B. 27
- C. 29
- D. 31

12. If $x : y = 2 : 3$ and $y : z = 4 : 5$, find $x : z$.

- A. 8 : 15
- B. 2 : 5
- C. 6 : 10
- D. 8 : 12

13. Factorize completely: $4a^2 - 9b^2$.

- A. $(2a - 3b)(2a - 3b)$
- B. $(2a + 3b)(2a - 3b)$
- C. $(4a - 9b)(a + b)$
- D. $(2a + 9b)(2a - b)$

14. A bag contains 5 red balls and 7 blue balls. What is the probability of picking a red ball at random?

- A. $\frac{5}{12}$
- B. $\frac{7}{12}$
- C. $\frac{5}{7}$
- D. $\frac{1}{5}$

15. Find the circumference of a circle whose diameter is 14 cm. [Take $\pi = \frac{22}{7}$]

- A. 22 cm
- B. 44 cm
- C. 88 cm
- D. 154 cm

16. If vector $u = (3, -2)$ and vector $v = (-1, 5)$, find $u + 2v$.

- A. (1, 8)
- B. (1, 3)
- C. (2, 8)
- D. (5, -4)

17. Solve the inequality: $2x - 5 \leq 3x + 1$.

- A. $x \geq -6$
- B. $x \leq -6$
- C. $x \geq 6$
- D. $x \leq 6$

18. Find the gradient (slope) of the line passing through points A(2, 3) and B(4, 7).

- A. $\frac{1}{2}$
- B. 1
- C. 2
- D. 4

19. A trader bought a pair of shoes for GHS 80 and sold it for GHS 100. Calculate the percentage profit.

- A. 20%
- B. 25%
- C. 30%
- D. 35%

20. If a car travels at an average speed of 60 km/h, what distance will it cover in 2 hours 30 minutes?

- A. 120 km
- B. 140 km
- C. 150 km
- D. 180 km

21. Simplify: $(x^2 - y^2) / (x - y)$.

- A. $x - y$
- B. $x + y$
- C. xy
- D. $x^2 + y^2$

22. The perimeter of a rectangle is 28 cm. If its length is 8 cm, find its width.

- A. 6 cm
- B. 8 cm
- C. 10 cm
- D. 12 cm

23. Find the value of y if $2^y = 32$.

- A. 4
- B. 5
- C. 6
- D. 7

24. Calculate the area of a right-angled triangle with base 6 cm and hypotenuse 10 cm.

- A. 24 cm^2
- B. 30 cm^2
- C. 48 cm^2
- D. 60 cm^2

25. Express 45% as a fraction in its lowest term.

- A. $\frac{9}{20}$
- B. $\frac{4}{10}$
- C. $\frac{9}{50}$
- D. $\frac{45}{100}$

26. Find the volume of a cylinder with radius 7 cm and height 10 cm. [Take $\pi = \frac{22}{7}$]

- A. 770 cm^3
- B. 1540 cm^3
- C. 2200 cm^3
- D. 3080 cm^3

27. What is the sum of angles in a quadrilateral?

- A. 180 deg
- B. 270 deg
- C. 360 deg
- D. 540 deg

28. Evaluate: 1.25×0.4 .

- A. 0.05
- B. 0.5
- C. 5.0
- D. 50.0

29. Make t the subject of the formula: $v = u + at$.

- A. $t = (v - u) / a$
- B. $t = (v + u) / a$
- C. $t = v / (u + a)$
- D. $t = a(v - u)$

30. If $3x + 2y = 12$ and $x = 2$, find the value of y.

- A. 2
- B. 3
- C. 4
- D. 6

31. Find the mean of the numbers: 12, 15, 18, 20, 25.
 A. 17
 B. 18
 C. 19
 D. 20
32. A map has a scale of 1:50,000. What actual distance is represented by 4 cm on the map?
 A. 1 km
 B. 2 km
 C. 2.5 km
 D. 5 km
33. Find the value of 5! (factorial of 5).
 A. 25
 B. 60
 C. 120
 D. 125
34. If an article is sold at a loss of 15% for GHS 170, find the cost price.
 A. GHS 190
 B. GHS 200
 C. GHS 210
 D. GHS 220
35. Which of the following is a prime number?
 A. 27
 B. 37
 C. 49
 D. 51
36. Find the size of an angle that is twice its supplement.
 A. 60 deg
 B. 90 deg
 C. 120 deg
 D. 150 deg
37. Find the value of x if $\log_{10}(x) = 3$.
 A. 30
 B. 100
 C. 300
 D. 1000
38. The bearing of town P from Q is 065 deg. What is the bearing of Q from P?
 A. 115 deg
 B. 205 deg
 C. 245 deg
 D. 335 deg
39. Calculate the total surface area of a cube with edge 4 cm.
 A. 64 cm^2
 B. 96 cm^2
 C. 120 cm^2
 D. 144 cm^2
40. Find the value of $(\frac{2}{3})^{-2}$.
 A. $\frac{4}{9}$
 B. $\frac{9}{4}$
 C. $-\frac{4}{9}$
 D. $-\frac{9}{4}$

OFFICIAL WAEC ANSWER KEY & SCORING GUIDE

For candidate self-assessment and teacher evaluation:

1. [A] 2. [C] 3. [A] 4. [B] 5. [A]
 6. [costly] 7. [C] 8. [C] 9. [C] 10. [B]
 11. [B] 12. [A] 13. [B] 14. [A] 15. [B]

16. [A] 17. [A] 18. [C] 19. [B] 20. [C]
21. [B] 22. [A] 23. [B] 24. [A] 25. [A]
26. [B] 27. [C] 28. [B] 29. [A] 30. [B]
31. [B] 32. [B] 33. [C] 34. [B] 35. [B]
36. [C] 37. [D] 38. [C] 39. [B] 40. [B]